

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

```
df = pd.read_csv("new.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
```

```

7   Parch      891 non-null   int64
8   Ticket     891 non-null   object
9   Fare       891 non-null   float64
10  Cabin      204 non-null   object
11  Embarked   889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

```
df['Survived'].value_counts()
```

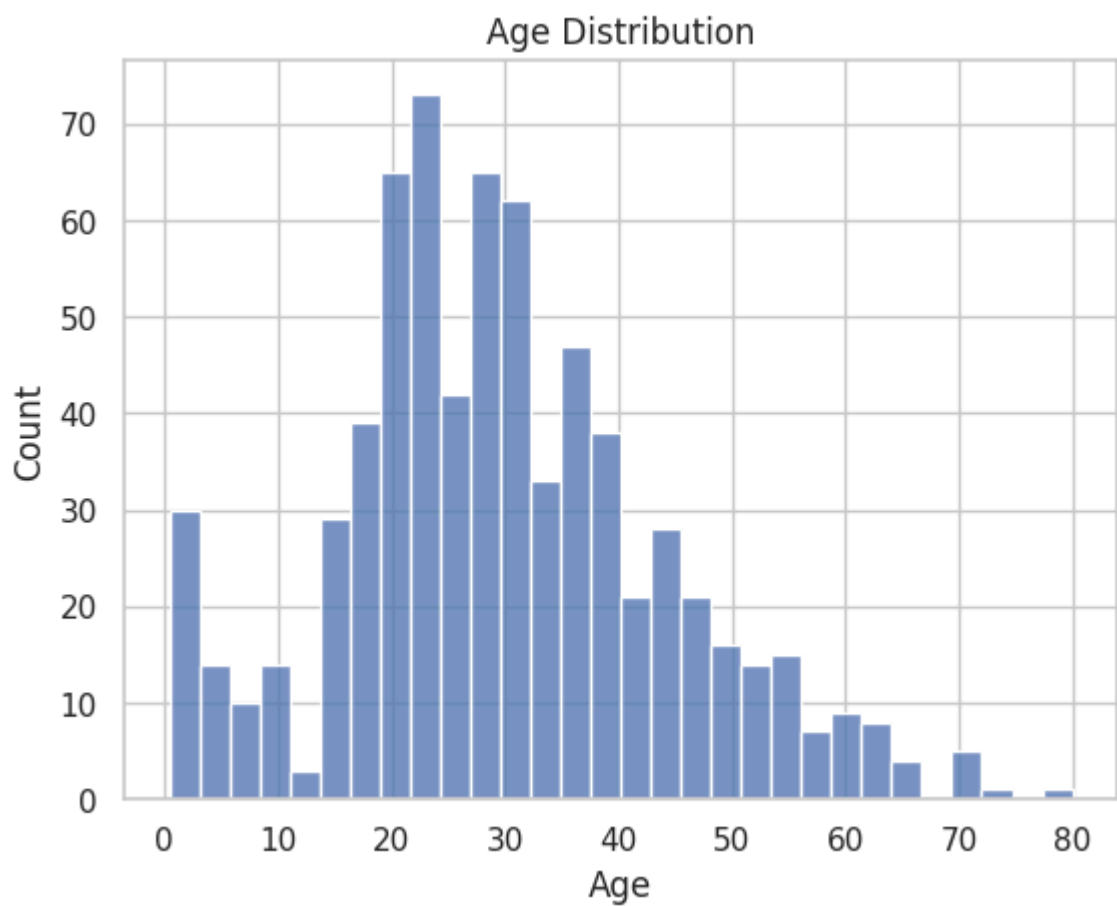
	count
Survived	
0	549
1	342

```
dtype: int64
```

```

plt.figure()
sns.histplot(df['Age'], bins=30)
plt.title("Age Distribution")
plt.show()

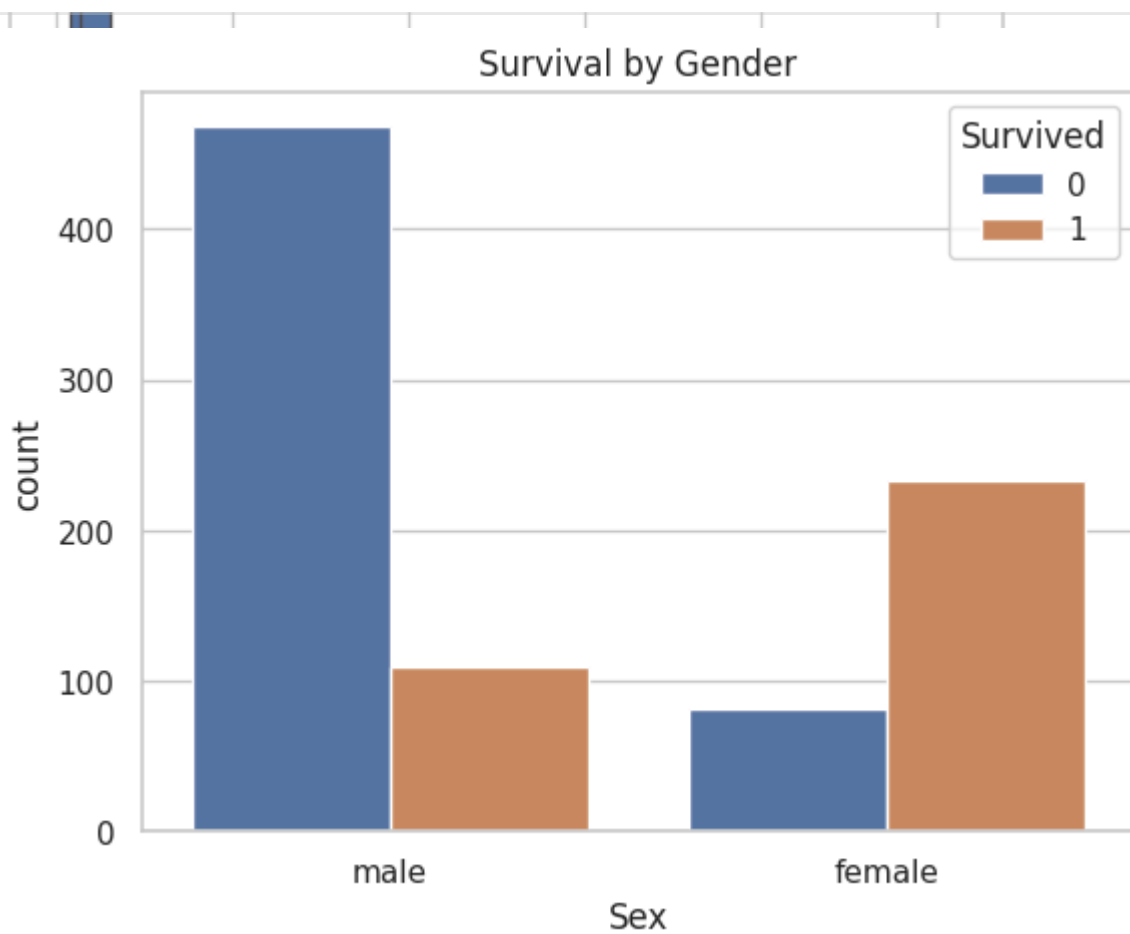
```



```
plt.figure()
sns.boxplot(x=df['Fare'])
plt.title("Fare Distribution")
plt.show()
```

Fare Distribution

```
plt.figure()
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()
```

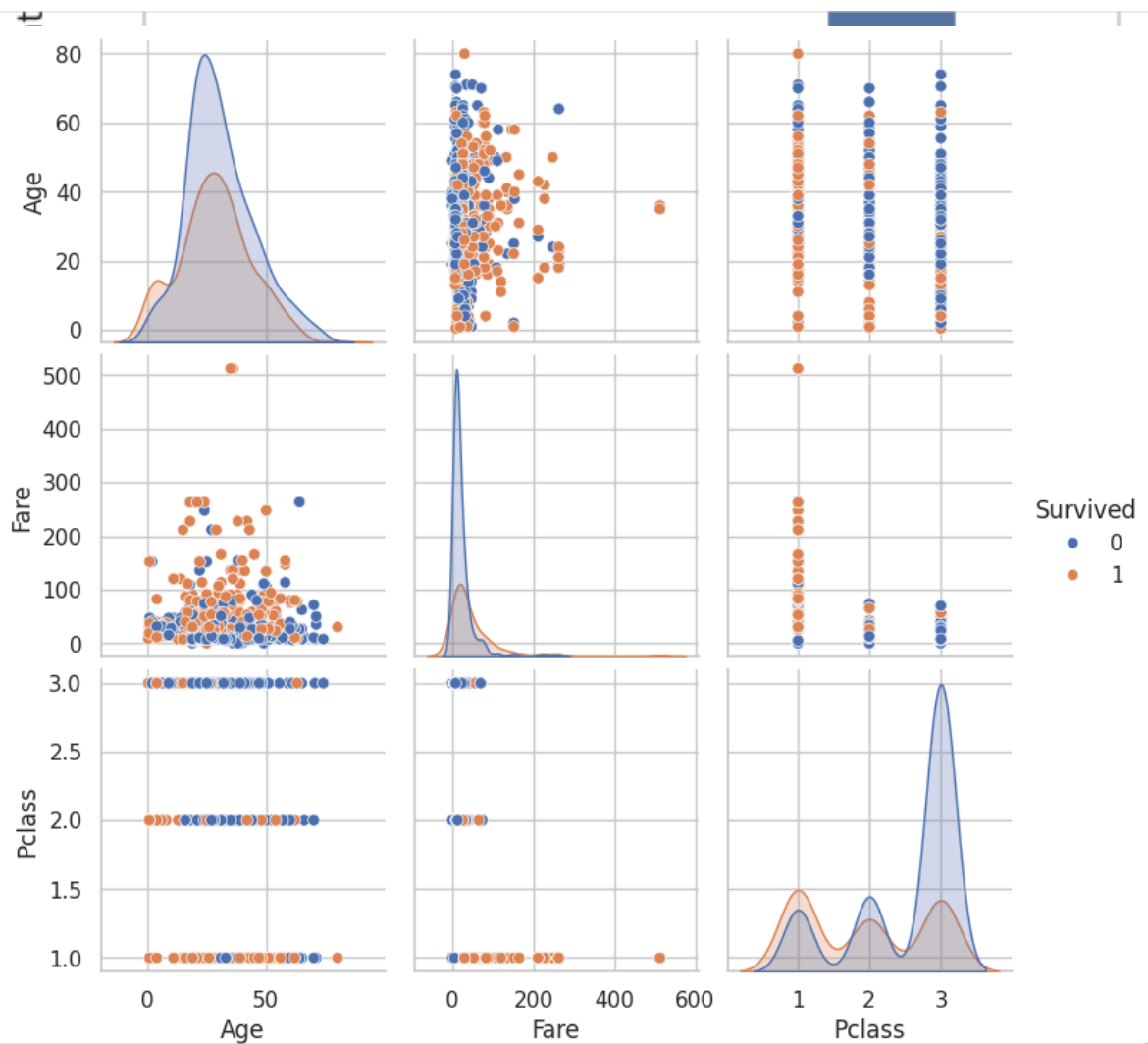


```
plt.figure()
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.show()
```

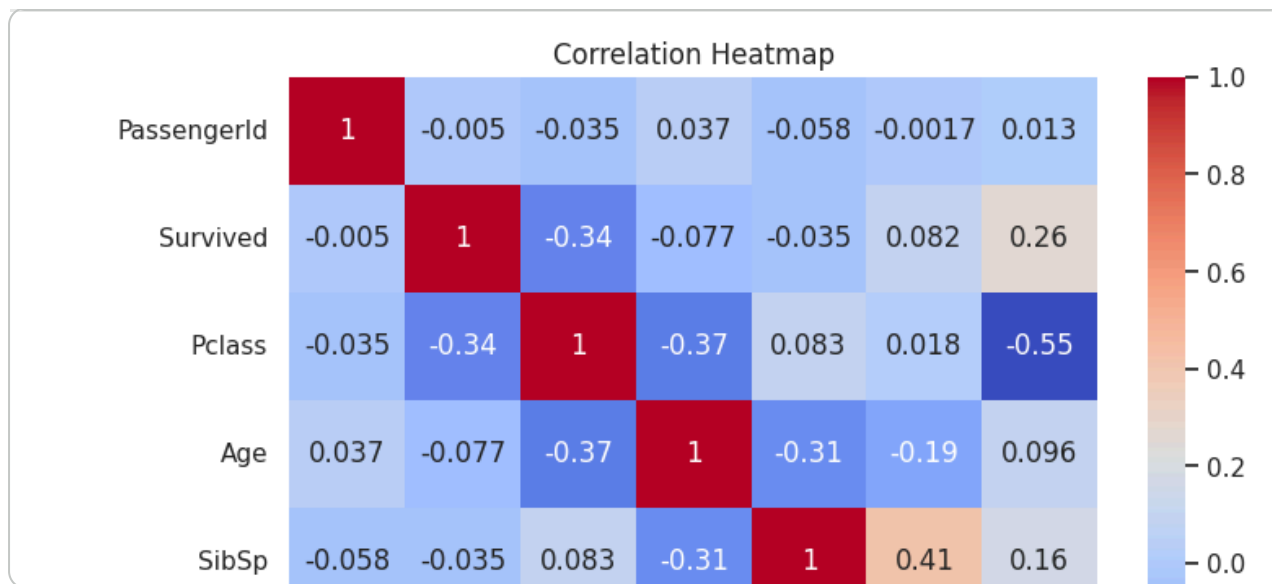
Survival by Passenger Class



```
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']], hue='Survived')
plt.show()
```



```
plt.figure(figsize=(8,6))
sns.heatmap(df.select_dtypes(include=['number']).corr(), annot=True, cmi :
plt.title("Correlation Heatmap")
plt.show()
```



```
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
print('Null values after imputation:')
print(df[['Age', 'Embarked']].isnull().sum())
```

```
Null values after imputation:
Age      0
Embarked 0
dtype: int64
```